

Coercivity of Hermitian and Toeplitz Form Matrices

A full positive answer to the structured-matrix question in arXiv:0807.2362

Automated Mathematics Run `fa_banach_001`

Lane 11, model GPT5.6

9 August 2026

Status: candidate full solution, likely valid. For bounded Hermitian block forms, scalar Schur-complement arguments transfer verbatim after Riesz representation. For bilateral and one-sided Toeplitz form matrices with absolutely summable operator coefficients, coercivity is equivalent to uniform positivity of the Hermitian part of the Fourier symbol; the theorem also gives the exact optimal coercivity constant.

1 Source question

The source is Stefano Cardanobile, *Diffusion systems and heat equations on networks* [1], arXiv:0807.2362 (2008). In the discussion of Section 1.3, printed page 49 (PDF page 60), it asks:

“An open question is whether it is possible to translate other arguments holding for scalar-valued matrices in the context of forms, e.g. symmetric matrices or Toeplitz matrices.”

developed for systems can be applied in this case. See also Example 1.11.

We also mention that it is possible to generalise the theory developed in Section 1.10 substituting the operator M_C^d by a more general operator of the form $\text{id} \otimes (M_{C_1}, \dots, M_{C_d})$.

Section 1.3

The results in this section are unpublished. The idea of the proof of Proposition 1.7 is due to Wolfgang Arendt. In Section 1.2 we have investigated coercivity properties for matrices in the case that the single forms have a particular expression. Proposition 1.7 is an example of a characterisation exploiting a particular structure of the form matrix. An open question is whether it is possible to translate other arguments holding for scalar-valued matrices in the context of forms, e.g. symmetric matrices or Toeplitz matrices.

Section 1.4

Proposition 2.0b) is due to Felix Ali-Mehmeti and Serge Nicaise in [44]. We slightly simplified the proof. Proposition 2.2 is already contained in [17]. Example 2.1 uses Ulm’s recipe for the investigation of multiplicative perturbations.

Figure 1: The open question in the original source, printed page 49. The full readable page width is retained.

The wording is existential and deliberately broad. The results below answer both named examples: an exact Schur criterion for Hermitian form matrices and an exact symbol criterion for a natural infinite-dimensional Toeplitz class.

2 Definitions and scope

All Hilbert spaces are complex and inner products are linear in the first variable. A bounded sesquilinear form a on a Hilbert space V has a unique Riesz representative $A \in \mathcal{B}(V)$ such that

$$a(u, v) = \langle Au, v \rangle.$$

Its optimal real lower bound is

$$c(a) := \inf_{u \neq 0} \frac{\operatorname{Re} a(u, u)}{\|u\|^2}.$$

Thus a is coercive exactly when $c(a) > 0$. The form is Hermitian when $a(u, v) = \overline{a(v, u)}$, equivalently $A = A^*$.

For the Toeplitz result, let $\mathcal{K}$ be an arbitrary complex Hilbert space and let $(B_k)_{k \in \mathbb{Z}} \subset \mathcal{B}(\mathcal{K})$ satisfy

$$\sum_{k \in \mathbb{Z}} \|B_k\| < \infty. \quad (1)$$

This includes every finite-band operator-valued Toeplitz matrix. For $J = \mathbb{Z}$ or $J = \mathbb{N}_0$, first define on finitely supported vectors $u, v \in \ell^2(J; \mathcal{K})$

$$a_J(u, v) := \sum_{m, n \in J} \langle B_{m-n} u_n, v_m \rangle, \quad (2)$$

and then extend continuously. Condition (1) guarantees this extension. Its norm-continuous symbol and Hermitian symbol are

$$F(\theta) := \sum_{k \in \mathbb{Z}} B_k e^{ik\theta}, \quad G(\theta) := \frac{F(\theta) + F(\theta)^*}{2}. \quad (3)$$

3 Proof intuition

For a Hermitian block form, Riesz representation changes the entries from scalar-valued sesquilinear maps into bounded operators. Gaussian elimination then becomes a bounded triangular congruence. The same multiplication that produces a scalar Schur complement produces an operator Schur complement, and bounded invertibility of the triangular factor makes strict positivity equivalent on both sides.

For Toeplitz forms, the repeated diagonal structure is convolution. Fourier series turn convolution into pointwise multiplication by $F(\theta)$, so the real quadratic form only sees $G(\theta)$. Uniform positivity of G is immediately sufficient. The only delicate direction is necessity for the one-sided matrix: normalized geometric sequences in $\ell^2(\mathbb{N}_0; \mathcal{K})$ have Fourier mass equal to a Poisson kernel. Concentrating that mass at an arbitrary angle and in an arbitrary coefficient-space direction recovers every pointwise Rayleigh quotient of G . No boundary loss occurs.

4 Symmetric matrices: the exact Schur criterion

Theorem 1 (Hermitian two-block form criterion). *Let V_1, V_2 be Hilbert spaces and let a bounded Hermitian form on $V_1 \oplus V_2$ have Riesz representative*

$$A = \begin{pmatrix} P & B^* \\ B & Q \end{pmatrix},$$

where $P = P^$ on V_1 , $Q = Q^*$ on V_2 , and $B \in \mathcal{B}(V_1, V_2)$. Suppose P is coercive. Define*

$$S := Q - BP^{-1}B^*.$$

Then the full form is coercive if and only if S is coercive.

Proof. Since P is coercive, P^{-1} is bounded. Direct block multiplication gives the exact factorization

$$A = \underbrace{\begin{pmatrix} I & 0 \\ BP^{-1} & I \end{pmatrix}}_L \begin{pmatrix} P & 0 \\ 0 & S \end{pmatrix} \underbrace{\begin{pmatrix} I & P^{-1}B^* \\ 0 & I \end{pmatrix}}_{L^*}. \quad (4)$$

The triangular operator L is boundedly invertible. If $P \geq pI$ and $S \geq sI$ with $p, s > 0$, then for every $z \in V_1 \oplus V_2$,

$$\langle Az, z \rangle \geq \min(p, s) \|L^*z\|^2 \geq \frac{\min(p, s)}{\|L^{-1}\|^2} \|z\|^2.$$

Thus A is coercive.

Conversely, suppose $A \geq \gamma I$ with $\gamma > 0$. Restricting to $V_1 \oplus \{0\}$ gives $P \geq \gamma I$. For $y \in V_2$, insert $z = (-P^{-1}B^*y, y)$ in the quadratic form. Formula (4) yields

$$\langle Sy, y \rangle = \langle Az, z \rangle \geq \gamma (\|P^{-1}B^*y\|^2 + \|y\|^2) \geq \gamma \|y\|^2.$$

Hence S is coercive. □

Corollary 2 (Finite Hermitian block elimination). *For a bounded Hermitian $n \times n$ form matrix on $V_1 \oplus \cdots \oplus V_n$, recursively take the first diagonal pivot and replace the remaining block by its Schur complement. The full form is coercive if and only if every pivot produced by this recursion is coercive.*

Proof. Apply Theorem 1 to the splitting $V_1 \oplus (V_2 \oplus \cdots \oplus V_n)$ and iterate on the Schur complement. If the full form is coercive, each leading compression and each resulting Schur complement is coercive by the converse part of the theorem. If every pivot is coercive, reverse the bounded triangular factorizations. □

This is precisely the operator/form version of Hermitian Gaussian elimination and the Schur-complement form of Sylvester's criterion.

5 Toeplitz matrices: exact symbol characterization

Theorem 3 (Bilateral and one-sided Toeplitz coercivity). *Assume (1), and let a_J be defined by (2), where $J = \mathbb{Z}$ or $J = \mathbb{N}_0$. Put*

$$g_* := \min_{\theta \in [0, 2\pi]} \inf_{\|x\|_{\mathcal{K}}=1} \langle G(\theta)x, x \rangle.$$

Then

$$c(a_J) = g_*. \quad (5)$$

In particular, the following are equivalent:

- (i) a_J is coercive;
- (ii) there is $\gamma > 0$ such that $G(\theta) \geq \gamma I_{\mathcal{K}}$ for every $\theta \in [0, 2\pi]$.

The same optimal constant works for the bilateral and one-sided forms.

Proof. Norm-uniform convergence in (3) makes F and G continuous. The function

$$h(\theta) := \inf_{\|x\|=1} \langle G(\theta)x, x \rangle$$

is continuous because $|h(\theta) - h(\varphi)| \leq \|G(\theta) - G(\varphi)\|$; hence g_* is a true minimum.

First consider finitely supported u, v and extend a one-sided vector by zero to $\mathbb{Z}$. With

$$(\mathcal{F}u)(\theta) := \sum_{n \in \mathbb{Z}} u_n e^{in\theta},$$

orthogonality of exponentials and the convention B_{m-n} give

$$a_J(u, v) = \frac{1}{2\pi} \int_0^{2\pi} \langle F(\theta)(\mathcal{F}u)(\theta), (\mathcal{F}v)(\theta) \rangle d\theta. \quad (6)$$

The absolute coefficient sum and vector-valued Young inequality show that the convolution operator is bounded on $\ell^2(\mathbb{Z}; \mathcal{K})$; the one-sided operator is a compression. Thus (6) extends to every $u, v \in \ell^2(J; \mathcal{K})$.

Taking real parts and using Parseval,

$$\begin{aligned} \operatorname{Re} a_J(u, u) &= \frac{1}{2\pi} \int_0^{2\pi} \langle G(\theta)(\mathcal{F}u)(\theta), (\mathcal{F}u)(\theta) \rangle d\theta \\ &\geq g_* \frac{1}{2\pi} \int_0^{2\pi} \|(\mathcal{F}u)(\theta)\|^2 d\theta = g_* \|u\|^2. \end{aligned}$$

Therefore $c(a_J) \geq g_*$.

For the reverse inequality fix $\theta_0 \in [0, 2\pi]$, a unit vector $x \in \mathcal{K}$, and $0 < r < 1$. In either index set use the vector supported on $\mathbb{N}_0$ given by

$$u_n^{(r)} = \sqrt{1 - r^2} r^n e^{-in\theta_0} x, \quad n \in \mathbb{N}_0.$$

It has norm one, and

$$\|\mathcal{F}u^{(r)}(\theta)\|^2 = \frac{1 - r^2}{|1 - re^{i(\theta - \theta_0)}|^2} =: P_r(\theta - \theta_0),$$

the Poisson kernel. Formula (6) gives

$$\operatorname{Re} a_J(u^{(r)}, u^{(r)}) = \frac{1}{2\pi} \int_0^{2\pi} P_r(\theta - \theta_0) \langle G(\theta)x, x \rangle d\theta.$$

The scalar function under the Poisson average is continuous, so the approximate-identity property yields, as $r \uparrow 1$,

$$\operatorname{Re} a_J(u^{(r)}, u^{(r)}) \longrightarrow \langle G(\theta_0)x, x \rangle.$$

Consequently $c(a_J) \leq \langle G(\theta_0)x, x \rangle$ for every θ_0 and every unit x . Taking the two infima gives $c(a_J) \leq g_*$. Combined with the first inequality this proves (5) and both equivalences. $\square$

Remark 4 (Why the one-sided case is not weaker). For a compression, pointwise symbol positivity is automatically sufficient, but necessity can fail for arbitrary compression problems. Here the Hardy geometric vectors concentrate at any chosen boundary point. They recover every pointwise Rayleigh quotient and therefore prevent an artificial gain in the coercivity constant.

6 Incidental correction to the source

The open question is preceded by Proposition 17 in the source. That proposition is not needed above and is false as written.

Proposition 5. *Condition (1.22) of source Proposition 17 is not necessary for coercivity, even when both component spaces are one-dimensional.*

Proof. Take $V_1 = V_2 = \mathbb{C}$,

$$a_{11}(z, w) = z\overline{w}, \quad a_{22}(z, w) = z\overline{w}, \quad a_{12} = a_{21} = 0.$$

Both optimal diagonal coercivity constants are $\alpha_1 = \alpha_2 = 1$. The full form satisfies

$$a((z_1, z_2), (z_1, z_2)) = |z_1|^2 + |z_2|^2,$$

so its optimal coercivity constant is one. If source condition (1.22) held for some $\alpha > 0$, setting $z_1 = 1$ and $z_2 = 0$ would give

$$0 \geq \alpha(|z_1|^2 + |z_2|^2) - 2|z_1||z_2| = \alpha,$$

a contradiction. □

The error is structural: independently rescaling the two coordinates changes the norm at which coercivity is tested. Optimal lower bounds for the diagonal forms do not convert that family of rescalings into an equivalence. A separate proof-gap packet records the exact source statement and counterexample.

7 Verification and computational checks

The proof was adversarially checked at the following points:

1. Multiplying the three factors in (4) recovers the block matrix with lower-right block Q .
2. The Fourier convention in (6) selects $k + n = m$, hence exactly the coefficient B_{m-n} .
3. The normalized geometric vector has norm one and squared Fourier norm P_r ; it is valid in both index sets because it is supported in $\mathbb{N}_0$.
4. Infinite-dimensional $\mathcal{K}$ causes no compactness issue: the infimum over unit vectors need not be attained, and the proof only takes an infimum. The outer minimum exists by operator-norm continuity.

The diagnostic script `code/check_coercivity.py` checks 100 random finite Schur factorizations, a 2×2 matrix-valued banded Toeplitz symbol against a 384-block one-sided compression, and the source counterexample. It reports

$$g_{\text{grid}} = 2.475765077, \quad \lambda_{\min}(T_{384}^{\text{herm}}) = 2.475784964,$$

consistent with finite compressions approaching the symbol edge from above. These computations are diagnostics only; no numerical statement is used in the proof.

8 Novelty search, limitations, and review status

A bounded check on 9 August 2026 searched the run indexes and arXiv-focused web results using the exact source wording and the phrases *operator-valued Toeplitz coercivity symbol*, *Toeplitz sesquilinear forms coercive*, and *block operator matrix Schur complement coercivity*. No later paper explicitly claiming to resolve Cardanobile’s question was found. General Toeplitz positivity and Schur-complement theory are classical; see, for example, [2]. Accordingly, this packet claims a new self-contained identification and proof relative to the source question, not priority for the ingredients or a comprehensive literature novelty claim.

The Toeplitz theorem assumes the natural Wiener condition (1). It therefore covers every finite-band form matrix and a large norm-continuous class, but does not claim to classify arbitrary

weakly specified Toeplitz arrays. Since the source asks whether such translations are possible and names the two structures rather than demanding maximal generality, the packet classifies the result as a full positive answer. A reviewer adopting the stronger historical interpretation may conservatively reclassify it as a substantial partial result.

Human-review recommendation. Send to human review. The main proof checks are the Toeplitz indexing, the Hardy/Poisson necessity argument, and the intended scope of the informal source question. The mathematical theorem itself has no conditional dependency.

References

- [1] S. Cardanobile, *Diffusion systems and heat equations on networks*, PhD thesis, arXiv:0807.2362, 2008.
- [2] A. Böttcher and B. Silbermann, *Analysis of Toeplitz Operators*, 2nd ed., Springer Monographs in Mathematics, Springer, 2006.